# Writing Research Papers

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**A practical guide to making a paper's argument clear.** Decide what the paper asks and what its evidence establishes. Then organize the paper around that answer and write the prose.

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**Abstract**

Most writing problems begin before the first sentence. The authors have not agreed on the paper's question or on the limits of their strongest claim. This guide starts there. It shows how to tie each claim to evidence and order the paper around its main result. It then treats figures and sentences before turning to LaTeX and revision. A reader should be able to find each claim and see why the evidence supports it.

## Contents

# 1 Start with the paper's question

A paper becomes difficult to write when its authors have not decided what the reader should learn. Polishing cannot settle that question. Before drafting, state the scientific problem and the paper's answer in ordinary language.

## 1.1 Write the paper's contract

Write four sentences before writing any section:

1. **Problem.** What practical or scientific problem does the paper address?
2. **Gap.** Why does the strongest current method or explanation leave that problem unresolved?
3. **Claim.** What does this paper establish?
4. **Boundary.** Where does the claim stop, and what important question remains open?

These sentences form a contract with the reader. The introduction may motivate the problem and the body may develop the details, but neither should quietly change the contract. The boundary sentence matters as much as the central claim. It prevents a result in one regime from becoming a claim about every regime. It also keeps a measured effect separate from the explanation proposed for it.

A useful one-sentence test is:

> We study *[problem]* because *[current limitation]*; we show *[central result]* under *[scope]*, while *[limitation]* remains open.

The finished sentence need not appear in the paper. Its purpose is to expose disagreement among the authors. If two plausible versions imply different experiments or different section orders, settle that disagreement before line-editing the introduction.

## 1.2 Map claims to evidence

List every claim that a reader must accept for the paper's conclusion to hold. Then assign each claim a visible piece of evidence. A theorem supports a formal statement under its assumptions. A controlled comparison supports a comparative claim. An ablation isolates a design choice. A mechanism study tests an explanation. A qualitative result can make a failure mode concrete, but it rarely establishes frequency on its own.

| **Claim** | **Evidence needed** | **Boundary to state** |
|---|---|---|
| Headline result | Theorem or controlled benchmark | Assumptions, tasks, metrics, and uncertainty |
| Mechanism | Diagnostic intervention or measured intermediate behavior | Alternative explanations that remain possible |
| Practical value | Cost, deployment result, or stress test | Resources and settings not yet tested |

This map should reveal weak links. A contribution with no evidence is a proposal, not a result. Evidence with no role in the central argument may still belong in the appendix, but it should not dictate the main paper. When a claim depends on a missing control, record the gap instead of drafting around it.

### 1.3 Choose the level of the contribution

Papers make different kinds of contributions. A method paper may show that a new procedure resolves a concrete tradeoff. A mechanism paper may overturn a popular explanation through controlled tests. A theory paper may identify a failure and prove when it disappears. Some papers combine these forms, but one usually carries the main argument.

Name that primary contribution early. It determines which evidence deserves space and what Figure 1 must show. It should also govern the title and abstract. A paper centered on explanation should not market itself as an algorithm paper because it happens to introduce a diagnostic model. A paper centered on a theorem should not bury the regime in which the theorem applies.

## 2 Build the argument before the prose

The section list should follow the logic of the evidence. A fixed outline can hide the paper's strongest result under routine setup or give equal space to claims of very different importance. Decide what the reader must understand first and which evidence establishes each step. Section titles and paragraph topics follow from that sequence.

### 2.1 Choose the structure from the evidence

The example papers use three recurring architectures.

**Empirical method papers.** *DPPO* [Ren et al., 2024] and *OGPO* [Patil et al., 2026] begin with an operational bottleneck, identify a tradeoff in current methods, and introduce a method that changes that tradeoff. Headline comparisons establish the main result. Later experiments test the proposed explanation. They also show where performance breaks and whether it survives deployment. This architecture works when performance is the central claim but mechanism and reliability determine whether the result is convincing.

**Empirical mechanism papers.** *Much Ado About Noising* [Pan et al., 2025] organizes the paper around competing explanations. Controlled tests first show which explanations do not account for the observed performance. A minimal replacement then isolates the components that do matter, and later experiments examine their mechanism. This architecture fits a paper whose contribution is a corrected understanding. The negative and positive findings must remain distinct: rejecting one explanation does not prove another.

**Theory-led papers.** *Action Chunking* [Zhang et al., 2025] starts from a failure in continuous-control behavior cloning and studies two interventions in different regimes. The formal results explain when each intervention is sufficient, while experiments test the predictions that matter outside the model. This architecture fits a paper that moves from a failure or impossibility to guarantees under stated conditions. The regime should appear before the guarantee, both in prose and in section titles.

Treat these as shapes. A new paper may need a different order. All four examples open with a concrete problem. Each then exposes a mismatch in current thinking and resolves it as far as the evidence allows. The test is whether each section resolves a question created by the previous one.

### 2.2 Order results by argumentative value

In an empirical paper, show the headline result before low-level ablations. The reader needs to know that the method changes the outcome before learning which regularizer or schedule matters. Mechanism studies should follow the performance claim unless mechanism is itself the main claim.

In a theory paper, present the motivating failure before the intervention. State assumptions before the result that uses them, but avoid a long block of notation that the reader will not need for several pages. Definitions are easier to absorb when they arrive close to the question they answer.

Related work can appear early or late. Place it early when the reader must distinguish the method from close alternatives before understanding the formulation. Place it after the main results when a literature survey would delay the paper's question and answer. In either case, organize related work by technical distinction. A chronological list of papers does not explain what remains unresolved.

### 2.3 Draft the evidence before the framing

The order of the paper need not be the order of writing. A reliable drafting sequence is:

1. settle the claim–evidence map;
2. draft the central figures, tables, theorem statements, and captions;
3. write the results that interpret that evidence;
4. write the setup and method needed to understand those results;
5. write the introduction once the contribution hierarchy is stable;
6. write the abstract and title last.

This sequence reduces a common source of churn. An early abstract often commits the paper to claims that the final evidence does not support. An early introduction can also turn provisional organization into a constraint. The abstract should summarize the paper that exists, not the paper the authors first imagined.

## 3 Give each section one job

A section should answer one question; a paragraph should make one inference. This rule is more useful than a target page count because it makes duplication visible. When two sections answer the same question, combine them or decide which one owns the claim. When a section answers no question, remove it or move its useful details to the appendix.

### 3.1 The introduction establishes the argument

Open on the concrete task and explain why current methods or theory fail on it. Narrow quickly to the exact failure. The reader should understand the tension before seeing the solution.

Then state the central insight in ordinary language. The method name and technical machinery can follow. A reader who does not yet know the notation should still be able to say what changed and why that change might matter.

Present contributions as supported claims:

> **Sample efficiency.** We compare the method with [baselines] under [controlled setting] and find [result] on [tasks or regimes].

The heading names the claim, while the sentence gives the comparison and evidence. "We introduce a new algorithm," "we run extensive experiments," and "we provide theory" describe work performed. They do not tell the reader what the work established.

Close the introduction with the paper's conceptual takeaway and its scope. Do not repeat the same contribution in several formats. Choose one primary signposting device and let the other elements add information.

### 3.2 Include only setup that a result needs

Preliminaries should define only what the first substantive result uses. Keep standard material short. Spend space on the assumption or distinction that later carries a claim, especially when two methods use similar notation but optimize different objects.

A method section should explain the conceptual move before presenting every implementation choice. Start from the limitation identified in the introduction. Show exactly what computation changes and why that change addresses the limitation. Then give the formal definition and practical variants.

A theory section should state the regime and result before a long derivation. Readers need to know why a theorem is being proved and what question it answers. Keep the main proof idea near the theorem when it carries scientific insight. Move routine algebra and complete technical cases to the appendix. Never promote an intuition to theorem language, and never omit a condition because it makes the statement less elegant.

### 3.3 Experiments answer named questions

Begin the experimental section with the questions the experiments will answer. Explain how each task and metric distinguishes possible answers. Then justify the baselines and state what is held fixed. A comparison cannot support a clean mechanism claim when the architecture, training objective, and initialization all change at once.

The usual order is:

1. evaluation regimes and controlled comparison rules;
2. headline performance;
3. stress tests or deployment;
4. mechanism studies;
5. ablations of design decisions;
6. efficiency, failures, and limitations.

Change this order when the scientific question requires it. A mechanism paper may need a controlled falsification before any broad benchmark. A paper about sample cost should report environment interactions before wall-clock convenience. The metric must match the claim.

State a result as a comparison. Name the metric and setting before offering an explanation. Separate a performance comparison from a mechanism test. "Method A outperforms Method B" and "component X causes the improvement" are different claims and need different evidence.

### 3.4 Related work explains distinctions

Group prior work by the technical choices that matter for the current paper. For each group, explain its aim and the assumption that separates it from the current paper. End with the point that remains unresolved. Cite the closest alternatives directly. Broad claims such as "few works consider this setting" require a careful search and a clear definition of the setting.

Related work should not become a second introduction. Its job is to place the paper's contribution accurately, including work that weakens the novelty claim. A narrow, correct distinction is more credible than a broad claim of being first.

### 3.5 Discussion states what follows and what does not

Use the discussion to interpret the evidence at the right level. State the regimes in which the result should change practice or understanding. Then name the limitation that most affects that conclusion. Untested environments may narrow the result's breadth; a strong assumption may change where it applies.

The conclusion should not introduce a new contribution. It can state the central result, its consequence, and the open question that now matters. A generic recap weakens the ending. End on the most precise supported implication or boundary.

### 3.6 The appendix should be usable

A long appendix benefits from a table of contents and an order designed for readers. Put complete proofs and the details needed for reproduction there. Practitioner guidance and qualitative failure analysis also belong in the appendix when they help readers reuse the work.

The main paper must still support its central claim. Do not move a decisive control or a condition that changes the theorem's meaning to the appendix merely to save space. Move details that permit verification and reuse while keeping the argument itself visible.

## 4 Build the argument into the figures

Readers often meet the evidence through figures before they read the full paper. The figure sequence should make the same argument in shorter form. A reader who scans the title, abstract, and figures should understand the problem and the main result. If the paper proposes an explanation, the figures should show the evidence for it.

### 4.1 Use Figure 1 to summarize the paper

Figure 1 should identify the new intervention and show its main effect. If the paper makes a mechanism claim, the figure should also show the evidence that motivates it. Most first figures pair a method diagram with one headline result. Add a task image or mechanism sketch only when it supplies necessary context.

The example papers make different choices because their claims differ. *Action Chunking* places its two interventions in one conceptual frame. *DPPO* joins the nested decision process to robot tasks and the properties it studies. *Much Ado About Noising* uses a component taxonomy to reject prevailing explanations and introduce a minimal replacement. *OGPO* places the algorithm beside sample-efficiency curves, then uses its final panel to motivate the mechanism study. In each case, the first figure reflects the paper's main claim rather than a standard figure template.

Design Figure 1 after the claim–evidence map is stable. If the figure requires a claim that the paper does not test, change the figure or collect the missing evidence. If the figure cannot state the contribution without several paragraphs of caption, the contribution hierarchy may still be unsettled.

## 4.2 Give later figures distinct roles

A later figure should resolve one question left open by Figure 1. Common roles include:

**Concept figure.**
Makes a theoretical object, information flow, or mechanism visible.

**Controlled comparison.**
Changes one disputed component while holding the relevant alternatives fixed.

**Headline benchmark.**
Establishes the main comparison across representative regimes.

**Mechanism figure.**
Shows trajectories, distributions, gradients, or intermediate computations that bear on an explanation.

**Qualitative sequence.**
Displays an ordered success or failure with the event of interest marked.

**Ablation.**
Removes or varies one design decision and reports the resulting change.

Do not ask every figure to perform all of these jobs. A benchmark grid becomes hard to read when it also contains the method diagram, ablation, and mechanism claim. Separate figures let the prose distinguish performance from explanation.

## 4.3 Use one visual language

Assign colors by meaning and keep those assignments stable. Use neutral gray for context and most baselines, then reserve one or two accent colors for the method and the main alternative. Learning curves should state the uncertainty shown and the number of seeds or trials. Direct labels are often easier to read than a distant legend. Panel headings should name the comparison, not merely "left" and "right."

Diagrams need a visible direction of travel. Give each arrow one meaning and state that meaning in the caption. Task images can ground an abstract diagram, but they should show the actual task or failure under discussion. A decorative image does not supply evidence.

Inspect every figure at its final column width. Check every label and annotation at that size. Lines must remain distinguishable on a printed page, including in grayscale and under common forms of color-vision deficiency.

## 4.4 Write captions as evidence summaries

Start the caption with the figure's claim or purpose. Walk through the panels in reading order. Then report the comparison and main observation before giving the narrowest interpretation the figure supports. Point to setup details when they are not visible nearby.

A useful caption order is:

1. claim or purpose;
2. panel-by-panel setup;
3. comparison and metric;
4. main observation;
5. narrow interpretation or caveat;
6. pointer to complete details.

The caption and results paragraph should agree. A caption cannot say that an ablation "demonstrates the mechanism" while the text calls it suggestive. Define nonstandard symbols and uncertainty either in the caption or in the adjacent prose. The reader should not need to search the appendix to learn what an error band means.

# 5 Write direct technical prose

Good technical prose lets the argument remain visible at sentence scale. Put the main subject and verb near the front. The example papers use first-person verbs to mark what the authors did. "We compare" names an experimental choice, while "we find" reports its result. They reserve "we hypothesize" for an interpretation. Use the technical object as the subject when the behavior belongs to the object.

> **Author action:** We compare the two methods under a shared initialization.
>
> **Technical behavior:** Action chunking limits compounding error under the stated stability condition.

Direct prose does not require short sentences. A long sentence works when every clause serves one relation. A short sentence fails when it adds drama but no claim.

## 5.1 Put conditions where they matter

A fronted clause should change how the main claim is understood:

> Although the estimator is biased, its error does not grow with the horizon in this regime.
>
> When the system is open-loop stable, action chunking gives the stated bound.
>
> Because the critic supplies the terminal reward, the method can reuse environment data off policy.

Openings such as "while" and "although" become clutter when they provide generic scene-setting. If the clause does not change how the main statement should be read, lead with the statement.

## 5.2 Make one main inference per sentence

A sentence may contain a result and its essential condition or qualification. Split it as soon as a second inference enters. The method, result, and proposed explanation rarely belong in the same sentence.

> OGPO reuses environment data through an off-policy critic. It spends additional computation on denoising trajectories instead of collecting new environment interactions.

The two sentences describe related parts of the same idea, but each has one main job. This separation also makes technical review easier. A coauthor can challenge the data-reuse claim without untangling three other claims from the same sentence.

### 5.3 Give each paragraph one inference

Most technical paragraphs make three or four moves:

1. state the claim;
2. give the reason or technical distinction;
3. present the theorem, comparison, measurement, or example;
4. state the supported consequence or boundary.

Use only the moves the paragraph needs. A two-sentence paragraph can be complete. Vary sentence length and syntax so that paragraphs do not read as filled-in forms. End on a consequence, limit, or transition that creates the next question. A final sentence that merely repeats the first can usually be deleted.

Paragraphs should also have explicit agents. Write "we compare" instead of "a comparison is conducted," and "the policy fails" instead of "a failure is exhibited by the policy." Nominalizations hide the action and lengthen the sentence. Direct verbs leave more room for the scientific detail.

### 5.4 Report numbers in a fixed logical order

For a quantitative result, lead with the comparator and metric. Then report the value under the exact setting, including uncertainty, before interpreting it:

> Compared with [baseline], [method] improves [metric] from [A] to [B] on [setting], averaged over [seeds or trials]. This supports [narrow inference].

Use the available numbers. "Substantially better" is less useful than the measured difference. Reserve "significant" for a defined statistical claim. If the paper reports uncertainty, say what it represents. If two methods use different budgets, data, or initializations, expose that difference before the reader reaches the result.

### 5.5 Separate formal results from interpretation

State the regime before a theorem's consequence:

> Under [assumptions], [intervention] guarantees [bound].

Then interpret the statement in a separate sentence:

> Theorem 2 bounds [quantity] by [rate]. Thus, in [regime], the error does not grow with [variable].

Preserve every quantifier and condition. Keep the asymptotic rate and scope exact. A clean sentence with a missing assumption is wrong. If an intuitive explanation is not part of the theorem, label it as intuition rather than folding it into the formal claim.

### 5.6 Calibrate mechanism language

Observation and explanation are different sentence types:

> We observe [measured behavior]. We hypothesize that [mechanism] produces this behavior because [support].

Use "shows" or "demonstrates" for direct evidence. Use "supports" for partial evidence. Use "suggests" or "is consistent with" for an interpretation that remains uncertain. "We believe" does not supply evidence for a hypothesis.

A controlled intervention can support a causal statement when the relevant alternatives are held fixed. A visualization can reveal structure without proving cause; the same is true of a correlation. The verb should make that distinction visible.

### 5.7 Use contrast to state scientific distinctions

Contrast is useful when it identifies a real scientific distinction. Sample efficiency differs from cheap computation; performance evidence differs from mechanism evidence. State both sides in one clear relation.

Repeated formulas such as "This is not X. It is Y" or "not only X but also Y" make the prose sound rehearsed. They also encourage false binaries. Write the scientific distinction directly:

> Current methods trade off sample efficiency and full-policy improvement.

When evidence rules out one explanation but does not prove another, say both facts without turning them into a slogan.

### 5.8 Replace abstract wrappers with verbs

Phrases such as "the mechanism behind X," "the role of Y in Z," and "the interplay between A and B" name a relationship without saying what happens. Use a direct verb when the evidence supports one:

> **Abstract:** We study the mechanisms behind *OGPO*'s stability.
>
> **Direct:** We test whether conservative advantages prevent critic exploitation.

Sometimes the relationship itself is the question. State that question precisely. "We test whether stochastic supervision causes manifold adherence" is clearer than "we study the role of stochasticity in manifold adherence," and it does not claim that the test has already succeeded.

### 5.9 Use lists only when the set matters

A sentence is not difficult because it is long. It becomes difficult when commas hold together several claims that have no hierarchy. The example papers often use long sentences to develop one contrast or causal relation. Follow that pattern.

Keep a full list when its members form a taxonomy, a set of assumptions, or a checklist. In ordinary prose, decide how the items relate. Group them by role or split the sentence:

> **Packed:** We evaluate stability, robustness, efficiency, and generalization.
>
> **Grouped:** We first measure training stability and sample efficiency. Separate deployment tests measure robustness and generalization.

### 5.10 Keep terms stable and use punctuation for grammar

Introduce a technical term once, then give its operational meaning. Repeat the canonical term instead of cycling through synonyms. Use a pronoun only when its referent is obvious. Semantic LaTeX macros help keep method names and symbols stable across authors and sections.

Use punctuation to expose the grammar. Commas mark real clause boundaries. Semicolons join closely related claims; colons introduce definitions or lists. Parentheses should contain secondary detail. Research questions belong in the prose when the question organizes the paper, not as a routine rhetorical setup. No sentence should depend on punctuation for drama.

## 6 Match every claim to the evidence

Technical editing begins with preservation. Keep every theorem condition and reported value intact. Preserve the notation, citations, and stated uncertainty. Do not make a sentence cleaner by deleting the fact that limits it.

### 6.1 Check the source before strengthening prose

Read the manuscript passage and the evidence it cites before editing a claim. Check a figure together with its caption, then consult the appendix for omitted conditions. A result that appears broad in the introduction may depend on a narrow metric or a specific initialization. A theorem may use a stronger assumption than the surrounding discussion.

Never invent evidence or imply that planned work is complete. Preserve every theorem condition and quantitative result. Mark a missing citation, fact, or control for the authors. Words cannot replace the missing evidence.

### 6.2 Trace each promise to evidence

Each abstract claim should appear in the main paper. Contributions in the introduction should point to visible evidence. A figure caption should make the same claim as the paragraph that interprets it, and the conclusion should follow from results already presented.

Run these checks in both directions. A promised contribution with no evidence is an overclaim. A decisive result that never appears in the abstract or introduction may mean the paper's contribution hierarchy is wrong. Repair the hierarchy before polishing sentences.

### 6.3 State comparison rules

State a comparison well enough that a reader can reconstruct its logic. Name the comparator, metric, and task. Then record any difference in budget, initialization, or data source. Mechanism claims require tighter controls than broad performance comparisons because they aim to isolate a cause.

Report failures and negative results with the same precision as successes. They can identify the regime boundary, reveal sensitivity, or reject an explanation. Do not hide them in vague limitation language.

### 6.4 Treat limitations as part of the result

A limitation is useful when it changes how the result should be applied or interpreted. Name the condition that creates the limit and say what remains supported. If several limits matter, explain each one separately.

Do not use a broad limitations paragraph to excuse a broad claim. Narrow the claim where it first appears. Readers should not need to reach the final page to learn that the central result holds only in one setting.

# 7 Keep the LaTeX source easy to reason about

The source tree determines how easily authors can revise the paper. A clear project prevents duplicated text and accidental notation changes. It also lets several authors work without breaking a submission mode.

## 7.1 Use the entry point as an outline

Keep `main.tex` short enough to read as the manuscript's table of contents. It should show the build mode and manuscript order at a glance. Put prose in section files:

```
paper/
|-- main.tex
|-- references.bib
|-- body/
|   |-- abstract.tex
|   |-- introduction.tex
|   |-- method.tex
|   `-- experiments.tex
|-- appendix/
|-- figures/
`-- preamble/
```

This guide preserves the complete preamble and title structure from OGPO [Patil et al., 2026] so that the original formatting remains inspectable. A new paper should keep the modular structure but remove unused project-specific commands. Source-specific macro sprawl makes later revisions harder to audit. Delete duplicate package loads and commented-out manuscripts.

## 7.2 Separate presentation from meaning

Stable package and layout choices belong in a style file. Put paper-specific notation in a commands file, including method names and operators. Keep author comments and colored revisions in a drafting file that can be disabled before release.

Keep venue `.sty` and `.cls` files unmodified. Load the venue style once, before project-specific styling, and record its version. Do not hide notation, spacing fixes, or author comments inside vendor code.

Use macros for semantic objects:

```
\newcommand{\method}{\textsc{Method}\xspace}
\newcommand{\state}{s}
\DeclareMathOperator*{\argmaxop}{arg\,max}
```

A method rename should require one edit. Avoid macros that conceal a whole sentence or arbitrary formatting. Source should remain understandable without opening several files for each line.

### 7.3 Give each toggle one responsibility

Build toggles are useful when a release changes presentation. Use them for anonymous and public author blocks or arXiv metadata. They can also control acknowledgments, appendix length, and venue-approved theorem styles. They should not select different claims or experimental values. Two build modes with different science will drift.

Keep toggles near the top of `main.tex`. Build every supported mode before release. The public and anonymous versions may look different, but they should make the same scientific claims.

### 7.4 Keep figure sources reproducible

Store final vector diagrams and plots as PDF when possible, and raster task images as PNG. Keep a documented generation path for assets produced from code. Use stable descriptive filenames and labels. Place the figure and its caption in the section that argues from it.

Do not tune layout with repeated negative `\vspace` commands while the content is moving. Settle the figure, caption, and section order first. Local spacing patches are difficult to maintain and often break under another venue style.

### 7.5 Compile as part of writing

Compile after each meaningful change. A practical loop is:

1. run `latexmk` with errors treated as failures;
2. inspect undefined references, citations, missing files, and boxes that exceed the margin;
3. render the PDF and inspect every page at final size;
4. repeat after changes to figures, tables, title blocks, or build modes.

Before submission, build from a clean directory. Remove TODOs, author comments, and revision colors. Treat missing references and undefined citations as errors. Confirm that the source bundle contains every figure and bibliography file. A successful compiler exit is necessary, but it does not catch a clipped label or an unreadable plot. It also cannot tell you that a page is empty or a title crosses the margin.

## 8 Use the included LaTeX template

The source for this guide retains the preamble used by OGPO [Patil et al., 2026]. It is a working paper archive, not a generic LaTeX package. That distinction matters. The template contains reusable theorem, reference, color, and editing machinery, but it also contains OGPO-specific algorithm names and notation. Keep the former. Replace the latter before starting a new paper.

Begin with a clean copy of the project. Replace the files in `body/` and `appendix/`, clear the bibliography, and update the title block in `main.tex`. Then remove unused definitions from `specific_macros.tex` and `algnames.tex`. A copied macro that never appears in the new paper is not harmless: it makes collisions and stale terminology harder to find.

## 8.1 Know which file owns each choice

The include order lives in `preamble/_preamble_includes.tex`. Keep that order until there is a specific reason to change it. The files have different jobs:

**`color_defs.tex`**
: Defines link colors, theorem colors, emphasis colors, and the `AIbox` environment. It loads before files that use those colors.

**`header.tex`**
: Loads common packages and defines page geometry, citations, cross references, algorithms, and the arXiv-dependent heading commands.

**`theorem_styles.tex`**
: Defines theorem-like environments and their appearance.

**`characters.tex` and `general_macros.tex`**
: Define reusable mathematical characters, delimiters, operators, and formatting commands.

**`specific_macros.tex`**
: Holds notation that belongs to one paper. This file should change substantially when the project changes.

**`algnames.tex`**
: Defines method names, method colors, task names, and the algorithm font. Treat it as a paper-specific legend.

**`color-edits.sty` and `comment_macros.tex`**
: Define temporary author edits, comments, deletion marks, and drafting flags.

**`arxiv_title.tex`**
: Defines the public title layout and metadata commands for websites, code, documentation, models, and dates.

The preamble is easiest to maintain when each new command goes into the narrowest file that owns it. A task name does not belong in `header.tex`. A package import does not belong in `specific_macros.tex`.

## 8.2 Use the theorem environments that are actually defined

With `customthms` enabled, choose an environment by its role:

- Use `theorem`, `lemma`, `corollary`, `proposition`, or `claim` for results.
- Use `definition`, `assumption`, or `condition` for definitions and conditions.
- Use `remark` or `example` for commentary and examples.

The template defines several less common environments as well. Use the full names `definition` and `theorem`; it does not define `defn` or `thm`.

```
\begin{definition}[Stable rollout]\label{def:stable-rollout}
A rollout is stable if ...
\end{definition}

\begin{theorem}[Main guarantee]\label{thm:main}
Under \Cref{def:stable-rollout}, ...
\end{theorem}

By \Cref{thm:main}, the error ...
```

Put the label immediately after the opening environment. Use `\Cref` at the beginning of a sentence and `\cref` within one. The template already names equations, assumptions, conditions, and components for `cleveref`. Equations are numbered by section.

Starred forms such as `theorem*`, `definition*`, and `remark*` suppress numbering. Use them only when no later sentence will need to cite the statement. `thminformal` provides a numbered informal theorem. An informal statement should still match the assumptions and conclusion of its formal counterpart.

The template also contains helpers for lettered variants of an existing result. For example:

```
\begin{thmmod}{thm:main}{b}{Finite-sample form}
...
\end{thmmod}
```

This prints a theorem number based on `thm:main` with the suffix `b`. Related helpers include `lemmod`, `propmod`, `defnmod`, `asmmod`, `cormod`, and `condmod`. Use them only when the new statement is genuinely a named variant of an existing result. Several other inherited helpers assume definitions from the original project and are not a stable public interface. Compile a minimal example before relying on one.

When `customthms` is false, the template does not define the full set of environments. That branch assumes the venue style provides them. Do not turn the toggle off in a plain `article` build and expect an identical theorem system.

## 8.3 Choose build toggles by responsibility

The toggles appear near the top of `main.tex`:

```
\newtoggle{arxiv}
\toggletrue{arxiv}
\newtoggle{customthms}
\toggletrue{customthms}
```

`arxiv` controls the public title, geometry, fonts, page style, algorithm packages, and colored paragraph heads. The false branch is a hook for a venue build, not a complete venue style by itself. In particular, the template loads `algorithm` and `algorithmic` only in the arXiv branch. A conference style must supply compatible algorithm support before the same source will compile.

`customthms` controls the colored theorem definitions. The `toreturn` toggle controls whether return notes are visible; those notes use `\RETURNN`. The inherited `icml` and `colt` toggles should remain only if the new paper still has those build modes. Remove a dead toggle rather than leaving a branch no one compiles.

The arXiv title macros also exist only when the arXiv title file is loaded. Calls such as `\paperdate` and direct access to `\metadatalist` must therefore stay inside the public build or receive venue-side definitions. The same rule applies to `\fancyhead`, because `fancyhdr` is loaded in the arXiv branch.

### 8.4 Use paragraph and emphasis commands deliberately

Three commands make emphasis change with the build mode:

**`\colorpar{Heading.}`**
: Creates a paragraph heading in both modes. The heading is colored in the arXiv build and plain in the venue branch.

**`\togglepar{Heading.}`**
: Calls `colorpar` in the arXiv build, but becomes inline bold text in the venue branch. Use it for a compact run-in heading that can lose the paragraph command under a tight style.

**`\colorbold{Text}`**
: Produces colored bold text in the arXiv build and ordinary bold text in the venue branch.

```
\togglepar{Experimental setup.} We compare ...

\colorpar{Failure under distribution shift.}
The policy fails when ...

\colorbold{Main finding.} The intervention ...
```

The period belongs inside the heading argument. Do not use these commands to make every paragraph look important. A paragraph heading should expose a real change in role: setup, result, mechanism, boundary, or implication.

`\termbold` and `\takeawaybold` use fixed semantic colors rather than a build toggle. They are appropriate only when the chosen venue permits that color. The paper must remain understandable when all emphasis is printed in grayscale.

### 8.5 Define method names and colors once

`algnames.tex` pairs a typographic method name with a stable color. The base helpers are:

```
\newcommand{\algofont}[1]{%
  {\scalefont{1.0}{\texttt{\textbf{#1}}}}}
\newcommand{\algofonttitle}[1]{%
  {\scalefont{1.08}{\textbf{\texttt{#1}}}}}
```

For a new method, define the color, a neutral name, a colored prose name, and a title form:

```
\definecolor{methodcol}{HTML}{A60000}
\newcommand{\Methodcolor}{methodcol}
\newcommand{\Methodnc}{\algofont{METHOD}}
\newcommand{\Method}{{\color{\Methodcolor}\Methodnc}}
\newcommand{\Methodtitle}{%
  {\color{\Methodcolor}\algofonttitle{METHOD}}}
```

Use \Method{} in prose so the following space is not swallowed. Use the neutral form when color would confuse a comparison or when the venue requires black text. Reserve the title form for a title or compact display; its font is deliberately larger.

Keep the same method-to-color assignment in plots and diagrams. Store the hex value in the plotting code as well as the LaTeX source. Color should help a reader track a method, but the method name, line style, or marker must still identify it without color.

The inherited file contains several aliases for the same method. It does not apply the neutral, colored, and title forms uniformly. Do not copy that inconsistency into a new method. Pick one naming pattern and remove aliases that the paper does not use.

### 8.6 Keep colors semantic

`color_defs.tex` centralizes four different jobs. Link and citation colors use dark teal. Theorem headings use dark red. Term and takeaway macros use stronger red accents. Method colors live in `algnames.tex`.

Change the semantic definition instead of scattering raw color commands through the manuscript. For example, changing `\thmcolordark` should update every theorem heading. It should not require a search through theorem prose.

The inherited names are not all standard color names. Inspect the actual `\definecolor` value before reusing one. A familiar name can point to a project-specific hex code.

The environment named `AIbox` is an inherited `tcolorbox`; its name does not prescribe its content. It is used for compact summaries:

```
\begin{AIbox}{Core insight}
The critic supplies ...
\end{AIbox}
```

Use a box only when the same short statement should remain visible while the reader scans the paper. Do not place a second box inside it, and do not use a box to repeat a contribution already stated in a heading and figure.

### 8.7 Use color edits as temporary review state

`comment_macros.tex` registers each author:

```
\addauthor{ms}{magenta}
\addauthor{xy}{blue}
```

Each registration creates four commands:

**`\xyedit{new text}`**
: Shows the text in the author's color. With the `suppress` option, the text remains and loses its color.

**`\xycomment{question}`**
: Shows a bracketed author comment. The `suppress` option hides it.

**`\xymargincomment{question}`**
: Adds a colored marker and small comment at that point. Despite the inherited name, the current implementation does not place the full text in a conventional margin note.

**\xydelete{old text}**

Marks a deletion. The `showdeletions` option displays struck-out text; the default displays a marker and deletion note.

Set package options where `color-edits.sty` is loaded:

```
\usepackage[showdeletions]{preamble/color-edits}
% or, for a clean diagnostic build:
\usepackage[suppress]{preamble/color-edits}
```

A suppressed build is a diagnostic, not a final cleanup. It can hide comments, deletion markers, and return notes that still remain in the source. Before release, remove resolved edit commands and search for author prefixes, `\REF`, `\CITE`, `\TODO`, and `\RETURNNN`. `\ignore` removes its argument in every build and should not hold text that might need to return.

### 8.8 Use the algorithm environment with its matching package

The arXiv branch uses the older `algorithmic` command set. Its commands are uppercase:

```
\begin{algorithm}[t]
\caption{Method update}\label{alg:update}
\begin{algorithmic}[1]
\REQUIRE Initial policy and replay buffer
\FOR{each update}
  \STATE Sample a batch
  \STATE Update the critic
\ENDFOR
\end{algorithmic}
\end{algorithm}
```

`\algcomment` provides the blue monospace comments used in the OGPO pseudocode. Use it only for short operational annotations. A paragraph of explanation belongs in the caption or surrounding text.

Do not mix `algorithmic`, `algpseudocode`, and `algorithm2e` commands. They expose similar names with different semantics. If a venue supplies its own algorithm package, adapt the source once and compile both modes.

### 8.9 Use notation shortcuts without hiding the mathematics

`characters.tex` creates families of common symbols. Use `\bA` for a bold matrix and `\bx` for a bold vector. The same file provides `\bbR` for blackboard-bold $R$ and `\cF` for calligraphic $F$.

`general_macros.tex` defines paired delimiters. These include `\abs`, `\prn`, and `\nrm`. It also defines operators such as `\argmin` and `\argmax`.

These generated names are convenient but broad. Check for an existing command before adding a new one. Do not redefine a familiar LaTeX command merely to save one character. When a symbol has paper-specific meaning, define a semantic name in `specific_macros.tex` instead of relying on a visual alias throughout the manuscript.

### 8.10 Populate the public title metadata carefully

The public title file can collect metadata below the title card:

```
\paperwebsite{https://project.example}
\papercode{https://github.com/group/project}
\paperdocs{https://project.example/docs}
\papermodels{https://huggingface.co/group/project}
\paperdate{\today}
```

`papercode`, `paperblog`, and `papermodels` include logo files from `logos/`. Keep those assets in the source bundle or replace the commands with plain links. Do not advertise an empty repository, private artifact, or placeholder URL.

The title card prints the entries in the order the commands are called. Include only metadata that helps a reader verify, reproduce, or use the work. The venue build should not depend on these public-title commands unless its style defines them.

### 8.11 Audit the template before submission

Build the public and venue modes from a clean directory. For each mode, check:

1. every theorem environment exists and every cross-reference resolves;
2. paragraph heads have the intended spacing after the toggle changes;
3. algorithm floats compile under the selected package;
4. method names retain the same spelling and color assignment;
5. no author edit, deletion marker, return note, or placeholder remains;
6. public metadata points to released artifacts; and
7. the source bundle contains its logos, figures, bibliography, style files, and any venue-owned class or bibliography style.

Finally, remove unused packages and source-specific macros. The preserved OGPO preamble is useful because its decisions remain visible. It becomes a burden when a new paper carries all of those decisions without knowing which ones it still needs.

## 9 Revise in separate passes

A single revision pass rarely works. Argument, evidence, and prose require different kinds of attention. Revise from the largest decision to the smallest.

### 9.1 Pass 1: argument

Read only what a skimming reader sees. Start with the title and abstract, follow the section headings and figure captions, and finish with the conclusion. Write down the central claim implied by that path. Compare it with the paper contract from Section 1. If they differ, fix the claim hierarchy and section order before editing prose.

Check whether each section answers one question and whether the sequence of questions is necessary. Remove duplicate motivation and contribution lists. Move secondary material out of the path of the main result.

## 9.2 Pass 2: evidence

Trace each major claim to the evidence that supports it. Verify every number and citation against the source. Check assumptions and uncertainty separately. A mechanism statement needs mechanism evidence; a performance statement needs a fair comparison.

Record missing controls and unresolved logic plainly. This pass may weaken a sentence or narrow a contribution. That is a correction, not a stylistic loss.

## 9.3 Pass 3: sections and paragraphs

Give each paragraph one inferential job. Put setup near the result that needs it. Move implementation detail, complete proofs, and secondary sweeps to the appendix when the main argument remains self-contained.

Check transitions by reading the last sentence of one paragraph and the first sentence of the next. The first should create or answer a real question. Add a transition only when the logical connection is missing; do not add a generic recap.

## 9.4 Pass 4: sentences and voice

Make the minimum effective line edit. Keep the writer's vocabulary and uncertainty. Preserve the cadence when it carries the argument. Leave strong sentences alone. Untangle a long sentence when it carries several inferences, but keep it when its clauses form one clear relation.

Cut throat-clearing and inflated claims. Remove qualifiers that add no meaning; replace abstract importance with a concrete result. Prefer a direct verb to a weak verb phrase. Repeat the correct technical term instead of rotating synonyms for variety.

Watch for canned contrasts and rhetorical question-and-answer setups. Repeated paragraph shapes can sound equally mechanical. These habits flatten distinct arguments into the same rhythm. Preserve a precise contrast or short sentence when it carries real scientific or personal force.

## 9.5 Pass 5: the rendered paper

Read the compiled PDF alongside the source. Inspect every page at normal zoom. Check the figures and tables first, then inspect equations and links. Make sure headings do not strand a line at the bottom of a page and that floats appear near the text that interprets them.

Then read the paper aloud or at speaking pace. This catches buried subjects, repeated syntax, and sentences whose grammar works but whose logic does not. The final manuscript should sound like a researcher explaining the result to a sharp colleague.

Stop when another edit would change only the surface. The last pass should leave the paper easier to verify, not merely more uniform.

## References

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Sarvesh Patil, Mitsuhiko Nakamoto, Manan Agarwal, Shashwat Saxena, Jesse Zhang, Giri Anantharaman, Cleah Winston, Chaoyi Pan, Douglas Chen, Nai-Chieh Huang, Zeynep Temel, Oliver Kroemer, Sergey Levine, Abhishek Gupta, Hongkai Dai, Paarth Shah, and Max Simchowitz. OGPO: Sample efficient full-finetuning of generative control policies. *arXiv preprint arXiv:2605.03065*, 2026. URL `https://arxiv.org/abs/2605.03065v4`.

Allen Z. Ren, Justin Lidard, Lars L. Ankile, Anthony Simeonov, Pulkit Agrawal, Anirudha Majumdar, Benjamin Burchfiel, Hongkai Dai, and Max Simchowitz. Diffusion policy policy optimization. *arXiv preprint arXiv:2409.00588*, 2024. URL `https://arxiv.org/abs/2409.00588v3`.

Thomas T. Zhang, Daniel Pfrommer, Chaoyi Pan, Nikolai Matni, and Max Simchowitz. Action chunking and exploratory data collection yield exponential improvements in behavior cloning for continuous control. *arXiv preprint arXiv:2507.09061*, 2025. URL `https://arxiv.org/abs/2507.09061v5`.

# A Paper-planning sheet

Complete this sheet before drafting a new paper or restructuring an existing one. Keep each answer short enough that coauthors can disagree with it precisely.

## A.1 The paper contract

1. **Problem:** What practical or scientific problem does the paper address?

2. **Gap:** Why does the strongest current method or explanation leave the problem unresolved?

3. **Central claim:** What does the paper establish?

4. **Scope and limitation:** Under what conditions does the claim hold, and where does it stop?

5. **Reader consequence:** What should a careful reader understand or do differently after seeing the evidence?

## A.2 Claim–evidence map

| Claim | Evidence | Boundary or missing test |
|---|---|---|
| | | |

## A.3 Section map

Write one question for each planned section. Remove a section when its question is already answered elsewhere.

| Section | Question | Decisive evidence |
|---|---|---|
| Introduction | | |
| Setup or preliminaries | | |
| Main result or method | | |
| Experiments or theory | | |
| Mechanism or ablations | | |
| Discussion | | |

## A.4 Figure sequence

1. **Figure 1:** What is new, what changes, and why might it change?
2. **Headline evidence:** Which figure or table establishes the central result?
3. **Mechanism or theory:** Which visual or theorem resolves the next question?
4. **Boundary:** Which result shows a failure, sensitivity, or limit?

## A.5 Drafting order

☐ Claim–evidence map ☐ Figures and captions ☐ Results ☐ Setup and method ☐ Introduction ☐ Abstract and title

# B Final manuscript check

## B.1 Argument and evidence

- ☐ The problem, gap, central claim, and boundary are explicit.
- ☐ Every abstract claim appears in the main paper.
- ☐ Every introduction contribution points to visible evidence.
- ☐ Each mechanism claim has evidence that bears on mechanism.
- ☐ Comparators, metrics, settings, budgets, and uncertainty are named.
- ☐ Limitations narrow claims where those claims first appear.
- ☐ The conclusion introduces no new result.

## B.2 Structure and prose

- ☐ Each section answers one question.
- ☐ Each paragraph performs one inferential job.
- ☐ Main subjects and verbs are easy to find.
- ☐ Opening clauses supply a real condition, cause, or contrast.
- ☐ Quantitative statements report values and settings before interpretation.
- ☐ Evidence verbs match the strength of the evidence.
- ☐ Technical terms, method names, and notation remain consistent.
- ☐ The final edit preserves the authors' voice and calibrated uncertainty.

## B.3 Figures and source

- ☐ Figure 1 communicates the problem, contribution, and main evidence.
- ☐ Every later figure has one primary role.
- ☐ Captions and surrounding prose make the same claim.
- ☐ Labels, legends, and annotations are legible at final size.
- ☐ `main.tex` records the manuscript and appendix order clearly.
- ☐ Venue files are unmodified and every supported mode builds.
- ☐ TODOs, comments, revision colors, and undefined references are gone.
- ☐ The clean source bundle includes every figure and bibliography file.

### B.4 Rendered PDF

- ☐ Every page has been inspected at normal zoom.
- ☐ No text, table, equation, or figure crosses a margin.
- ☐ Floats appear near the prose that interprets them.
- ☐ Links, footnotes, page numbers, and appendix references work.
- ☐ The paper reads naturally at speaking pace.